CHAPTER 23

Credit Risk

The value-at-risk measure we covered in Chapter 21 and the Greek letters we studied in Chapter 18 are aimed at quantifying market risk. In this chapter we consider another important risk for financial institutions: credit risk. Most financial institutions devote considerable resources to the measurement and management of credit risk. Regulators have for many years required banks to keep capital to reflect the credit risks they are bearing. This capital is in addition to the capital, described in Business Snapshot 21.1, that is required for market risk.

Credit risk arises from the possibility that borrowers and counterparties in derivatives transactions may default. This chapter discusses a number of different approaches to estimating the probability that a company will default and explains the key difference between risk-neutral and real-world probabilities of default. It examines the nature of the credit risk in over-the-counter derivatives transactions and discusses the clauses derivatives dealers write into their contracts to reduce credit risk. It also covers default correlation, Gaussian copula models, and the estimation of credit value at risk.

Chapter 24 will discuss credit derivatives and show how ideas introduced in this chapter can be used to value these instruments.

23.1 CREDIT RATINGS

Rating agencies, such as Moody's, S&P, and Fitch, are in the business of providing ratings describing the creditworthiness of corporate bonds. The best rating assigned by Moody's is Aaa. Bonds with this rating are considered to have almost no chance of defaulting. The next best rating is Aa. Following that comes A, Baa, Ba, B, Caa, Ca, and C. Only bonds with ratings of Baa or above are considered to be *investment grade.* The S&P and Fitch ratings corresponding to Moody's Aaa, Aa, A, Baa, Ba, B, Caa, Ca, and C are AAA, AA, A, BBB, BB, B, CCC, CC, and C, respectively. To create finer rating measures, Moody's divides its Aa rating category into Aa1, Aa2, and Aa3, its A category into A1, A2, and A3, and so on. Similarly, S&P and Fitch divide their AA rating category into AA+, AA, and AA−, their A rating category into A+, A, and A−, and so on. Moody's Aaa category and the S&P/Fitch AAA category are not subdivided, nor usually are the two lowest rating categories.

23.2 HISTORICAL DEFAULT PROBABILITIES

Table 23.1 is typical of the data produced by rating agencies. It shows the default experience during a 20-year period of bonds that had a particular rating at the beginning of the period. For example, a bond with a credit rating of Baa has a 0.176% chance of defaulting by the end of the first year, a 0.494% chance of defaulting by the end of the second year, and so on. The probability of a bond defaulting during a particular year can be calculated from the table. For example, the probability that a bond initially rated Baa will default during the second year is $0.494 - 0.176 = 0.318\%$.

Table 23.1 shows that, for investment-grade bonds, the probability of default in a year tends to be an increasing function of time (e.g., the probabilities of an A-rated bond defaulting during years 0–5, 5–10, 10–15, and 15–20 are 0.717%, 1.329%, 1.526%, and 2.362%, respectively). This is because the bond issuer is initially considered to be creditworthy, and the more time that elapses, the greater the possibility that its financial health will decline. For bonds with a poor credit rating, the probability of default is often a decreasing function of time (e.g., the probabilities that a B-rated bond will default during years 0–5, 5–10, 10–15, and 15–20 are 25.895%, 18.482%, 11.721%, and 6.380%, respectively). The reason here is that, for a bond with a poor credit rating, the next year or two may be critical. The longer the issuer survives, the greater the chance that its financial health improves.

Hazard Rates

From Table 23.1 we can calculate the probability of a bond rated Caa or below defaulting during the third year as $38.682 - 29.384 = 9.298\%$. We will refer to this as the *unconditional default probability*. It is the probability of default during the third year as seen at time 0. The probability that the bond will survive until the end of year 2 is $100 - 29.384 = 70.616\%$. The probability that it will default during the third year conditional on no earlier default is therefore $0.09298/0.70616$, or 13.17%. Conditional default probabilities are referred to as *hazard rates* or *default intensities*.

The 13.17% we have just calculated is for a 1-year time period. Suppose instead that we consider a short time period of length Δt. The hazard rate $\lambda(t)$ at time t is then defined so that $\lambda(t)\,\Delta t$ is the probability of default between time t and $t + \Delta t$ conditional on no earlier default. If $V(t)$ is the cumulative probability of the company surviving to time t (i.e., no default by time t), the conditional probability of default between time t

Table 23.1 Average cumulative default rates (%), 1970–2009. *Source*: Moody's.

Term (years):	*1*	*2*	*3*	*4*	*5*	*7*	*10*	*15*	*20*
Aaa	0.000	0.012	0.012	0.037	0.105	0.245	0.497	0.927	1.102
Aa	0.022	0.059	0.091	0.159	0.234	0.384	0.542	1.150	2.465
A	0.051	0.165	0.341	0.520	0.717	1.179	2.046	3.572	5.934
Baa	0.176	0.494	0.912	1.404	1.926	2.996	4.851	8.751	12.327
Ba	1.166	3.186	5.583	8.123	10.397	14.318	19.964	29.703	37.173
B	4.546	10.426	16.188	21.256	25.895	34.473	44.377	56.098	62.478
Caa–C	17.723	29.384	38.682	46.094	52.286	59.771	71.376	77.545	80.211

and $t + \Delta t$ is $[V(t) - V(t + \Delta t)]/V(t)$. Since this equals $\lambda(t)\,\Delta t$, it follows that

$$V(t + \Delta t) - V(t) = -\lambda(t)V(t)\,\Delta t$$

Taking limits

$$\frac{dV(t)}{dt} = -\lambda(t)V(t)$$

from which

$$V(t) = e^{-\int_0^t \lambda(\tau)d\tau}$$

Defining $Q(t)$ as the probability of default by time t, so that $Q(t) = 1 - V(t)$, gives

$$Q(t) = 1 - e^{-\int_0^t \lambda(\tau)d\tau}$$

or

$$Q(t) = 1 - e^{-\bar{\lambda}(t)t} \tag{23.1}$$

where $\bar{\lambda}(t)$ is the average hazard rate (default intensity) between time 0 and time t.

23.3 RECOVERY RATES

When a company goes bankrupt, those that are owed money by the company file claims against the assets of the company.[1] Sometimes there is a reorganization in which these creditors agree to a partial payment of their claims. In other cases the assets are sold by the liquidator and the proceeds are used to meet the claims as far as possible. Some claims typically have priority over other claims and are met more fully.

The recovery rate for a bond is normally defined as the bond's market value a few days after a default, as a percent of its face value. Table 23.2 provides historical data on average recovery rates for different categories of bank loans and bonds in the United States. It shows that bank loans with a first lien on assets had the best average recovery rate, 65.6%. For bonds, the average recovery rate ranges from 49.8% for those that are

Table 23.2 Recovery rates on corporate bonds as a percentage of face value, 1982–2009. *Source*: Moody's.

Class	*Average recovery rate (%)*
First lien bank loan	65.6
Second lien bank loan	32.8
Senior unsecured bank loan	48.7
Senior secured bond	49.8
Senior unsecured bond	36.6
Senior subordinated bond	30.7
Subordinated bond	31.3
Junior subordinated bond	24.7

[1] In the United States, the claim made by a bond holder is the bond's face value plus accrued interest.

both senior to other lenders and secured to 24.7% for those that rank after other lenders with a security interest that is subordinate to other lenders.

Recovery rates are significantly negatively correlated with default rates.[2] This means that a bad year for the default rate is usually doubly bad because it is accompanied by a low recovery rate. For example, when the default rate for non-investment-grade bonds in a year is only 0.1%, the average recovery rate might be relatively high at 60%. When the default rate is relatively high at 3%, the average recovery rate might be only 35%.

23.4 ESTIMATING DEFAULT PROBABILITIES FROM BOND PRICES

The probability of default for a company can be estimated from the prices of bonds it has issued. The usual assumption is that the only reason a corporate bond sells for less than a similar risk-free bond is the possibility of default.[3]

Consider first an approximate calculation. Suppose that a bond yields 200 basis points more than a similar risk-free bond and that the expected recovery rate in the event of a default is 40%. The holder of a corporate bond must be expecting to lose 200 basis points (or 2% per year) from defaults. Given the recovery rate of 40%, this leads to an estimate of the probability of a default per year conditional on no earlier default of $0.02/(1-0.4)$, or 3.33%. In general,

$$\bar{\lambda} = \frac{s}{1-R} \tag{23.2}$$

where $\bar{\lambda}$ is the average hazard rate (default intensity) per year, s is the spread of the corporate bond yield over the risk-free rate, and R is the expected recovery rate.

A More Exact Calculation

For a more exact calculation, suppose that the corporate bond we have been considering lasts for 5 years, provides a coupon 6% per annum (paid semiannually) and that the yield on the corporate bond is 7% per annum (with continuous compounding). The yield on a similar risk-free bond is 5% (with continuous compounding). The yields imply that the price of the corporate bond is 95.34 and the price of the risk-free bond is 104.09. The expected loss from default over the 5-year life of the bond is therefore $104.09-95.34$, or \$8.75. Suppose that the unconditional probability of default per year (assumed to be the same each year) is Q. Table 23.3 calculates the expected loss from default in terms of Q on the assumption that defaults can happen at times 0.5, 1.5, 2.5, 3.5, and 4.5 years (immediately before coupon payment dates). Risk-free rates for all maturities are assumed to be 5% (with continuous compounding).

To illustrate the calculations, consider the 3.5-year row in Table 23.3. The expected value of the corporate bond at time 3.5 years (calculated using forward interest rates

[2] See E. I. Altman, B. Brady, A. Resti, and A. Sironi, "The Link between Default and Recovery Rates: Theory, Empirical Evidence, and Implications," *Journal of Business*, 78, 6 (2005), 2203–28. This is also discussed in publications by Moody's Investors Service.

[3] This assumption is not perfect. In practice the price of a corporate bond is affected by its liquidity. The lower the liquidity, the lower the price.

Table 23.3 Calculation of loss from default on a bond in terms of the default probabilities per year, Q. Notional principal = \$100.

Time (years)	*Default probability*	*Recovery amount (\$)*	*Risk-free value (\$)*	*Loss given default (\$)*	*Discount factor*	*PV of expected loss (\$)*
0.5	Q	40	106.73	66.73	0.9753	$65.08Q$
1.5	Q	40	105.97	65.97	0.9277	$61.20Q$
2.5	Q	40	105.17	65.17	0.8825	$57.52Q$
3.5	Q	40	104.34	64.34	0.8395	$54.01Q$
4.5	Q	40	103.46	63.46	0.7985	$50.67Q$
Total						$288.48Q$

and assuming no possibility of default) is

$$3 + 3e^{-0.05\times 0.5} + 3e^{-0.05\times 1.0} + 103e^{-0.05\times 1.5} = 104.34$$

Given the definition of recovery rates in the previous section, the amount recovered if there is a default is 40, so that the loss given default is 104.34 − 40, or \$64.34. The present value of this loss is 54.01. The expected loss is therefore $54.01Q$.

The total expected loss is $288.48Q$. Setting this equal to 8.75, we obtain a value for Q of 8.75/288.48, or 3.03%. The calculations we have given assume that the default probability is the same in each year and that defaults take place at just one time during the year. We can extend the calculations to assume that defaults can take place more frequently. Also, instead of assuming a constant unconditional probability of default we can assume a constant hazard rate (default intensity) or assume a particular pattern for the variation of default probabilities with time. With several bonds we can estimate several parameters describing the term structure of default probabilities. Suppose, for example, we have bonds maturing in 3, 5, 7, and 10 years. We could use the first bond to estimate a default probability per year for the first 3 years, the second bond to estimate default probability per year for years 4 and 5, the third bond to estimate a default probability for years 6 and 7, and the fourth bond to estimate a default probability for years 8, 9, and 10 (see Problems 23.13 and 23.27). This approach is analogous to the bootstrap procedure in Section 4.5 for calculating a zero-coupon yield curve.

The Risk-Free Rate

A key issue when bond prices are used to estimate default probabilities is the meaning of the terms "risk-free rate" and "risk-free bond." In equation (23.2), the spread s is the excess of the corporate bond yield over the yield on a similar risk-free bond. In Table 23.3, the risk-free value of the bond must be calculated using the risk-free discount rate. The benchmark risk-free rate that is usually used in quoting corporate bond yields is the yield on similar Treasury bonds. (For example, a bond trader might quote the yield on a particular corporate bond as being a spread of 250 basis points over Treasuries.)

As discussed in Section 4.1, traders usually use LIBOR/swap rates as proxies for risk-free rates when valuing derivatives. Traders also often use LIBOR/swap rates as risk-free rates when calculating default probabilities. For example, when they determine default probabilities from bond prices, the spread s in equation (23.2) is the spread of the bond yield over the LIBOR/swap rate. Also, the risk-free discount rates used in the calculations in Table 23.3 are LIBOR/swap zero rates.

Credit default swaps (which will be discussed in the next chapter) can be used to imply the risk-free rate assumed by traders. The implied rate appears to be approximately equal to the LIBOR/swap rate minus 10 basis points on average.[4] This estimate is plausible. As explained in Section 7.5, the credit risk in a swap rate is the credit risk from making a series of short-term loans to AA-rated counterparties and 10 basis points is a reasonable default risk premium for a AA-rated short-term instrument.

Asset Swaps

In practice, traders often use asset swap spreads as a way of extracting default probabilities from bond prices. This is because asset swap spreads provide a direct estimate of the spread of bond yields over the LIBOR/swap curve.

To explain how asset swaps work, consider the situation where an asset swap spread for a particular bond is quoted as 150 basis points. There are three possible situations:

1. The bond sells for its par value of 100. The swap then involves one side (company A) paying the coupon on the bond and the other side (company B) paying LIBOR plus 150 basis points. Note that it is the promised coupons that are exchanged. The exchanges take place regardless of whether the bond defaults.
2. The bond sells below its par value, say, for 95. The swap is then structured so that, in addition to the coupons, company A pays \$5 per \$100 of notional principal at the outset. Company B pays LIBOR plus 150 basis points.
3. The underlying bond sells above par, say, for 108. The swap is then structured so that, in addition to LIBOR plus 150 basis points, company B makes a payment of \$8 per \$100 of principal at the outset. Company A pays the coupons.

The effect of all this is that the present value of the asset swap spread is the amount by which the price of the corporate bond is exceeded by the price of a similar risk-free bond where the risk-free rate is assumed to be given by the LIBOR/swap curve (see Problem 23.22).

Consider again the example in Table 23.3 where the LIBOR/swap zero curve is flat at 5%. Suppose that instead of knowing the bond's price we know that the asset swap spread is 150 basis points. This means that the amount by which the value of the risk-free bond exceeds the value of the corporate bond is the present value of 150 basis points per year for 5 years. Assuming semiannual payments, this is \$6.55 per \$100 of principal. The total loss in Table 23.3 would in this case be set equal to \$6.55. This means that the default probability per year, Q, would be 6.55/288.48, or 2.27%.

23.5 COMPARISON OF DEFAULT PROBABILITY ESTIMATES

The default probabilities estimated from historical data are usually much less than those derived from bond prices. The difference between the two was particularly large during

[4] See J. Hull, M. Predescu, and A. White, "The Relationship between Credit Default Swap Spreads, Bond Yields, and Credit Rating Announcements," *Journal of Banking and Finance*, 28 (November 2004): 2789-2811.

the credit crisis which started in mid-2007. This is because there was what is termed a "flight to quality" during the crisis, where all investors wanted to hold safe securities such as Treasury bonds. The prices of corporate bonds declined, thereby increasing their yields. The credit spread s on these bonds increased and calculations such as the one in equation (23.2) gave very high default probability estimates.

We now show that it was also true that default probabilities calculated from bonds were higher than those calculated before the credit crisis. We first calculate the historical default probabilities using the data in the 7-year column of Table 23.1. (We use the 7-year column because the bonds we will look at later have a life of about 7 years.) From equation (23.1), we have

$$\bar{\lambda}(7) = -\tfrac{1}{7}\ln[1 - Q(7)]$$

where $\bar{\lambda}(t)$ is the average hazard rate (or default intensity) by time t and $Q(t)$ is the cumulative probability of default by time t. The values of $Q(7)$ for different rating categories are in Table 23.1. For example, for an A-rated company, $Q(7)$ is 0.01179. The average 7-year hazard rate is therefore

$$\bar{\lambda}(7) = -\tfrac{1}{7}\ln(1 - 0.01179) = 0.0017$$

or 0.17%.

To calculate average hazard rates from bond prices, we use equation (23.2) and bond yields published by Merrill Lynch. The results shown are averages between December 1996 and June 2007. The recovery rate is assumed to be 40%. The Merrill Lynch bonds have a life of about seven years. (This explains why we focused on the 7-year column in Table 23.1 when calculating historical default probabilities.) To calculate the bond yield spread, we assume, to be consistent with the arguments in the previous section, that the risk-free interest rate is the 7-year swap rate minus 10 basis points. For example, for A-rated bonds, the average Merrill Lynch yield was 5.995%. The average 7-year swap rate was 5.408%, so that the average risk-free rate was 5.308%. This gives the average 7-year hazard rate as

$$\frac{0.05995 - 0.05308}{1 - 0.4} = 0.0115$$

or 1.15%.

Table 23.4 shows that the ratio of the hazard rate backed out from bond prices to the hazard rate calculated from historical data is very high for investment-grade companies

Table 23.4 Seven-year average hazard rates (% per annum).

Rating	*Historical hazard rate*	*Hazard rate from bonds*	*Ratio*	*Difference*
Aaa	0.04	0.60	17.0	0.56
Aa	0.05	0.73	13.2	0.67
A	0.17	1.15	6.8	0.98
Baa	0.43	2.13	4.9	1.69
Ba	2.21	4.67	2.1	2.46
B	6.04	8.02	1.3	1.98
Caa and lower	13.01	18.39	1.4	5.39

Table 23.5 Expected excess return on bonds (basis points).

Rating	*Bond yield spread over Treasuries*	*Spread of risk-free rate over Treasuries*	*Spread for historical defaults*	*Excess return*
Aaa	78	42	2	34
Aa	86	42	3	41
A	111	42	10	59
Baa	169	42	26	101
Ba	322	42	132	148
B	523	42	362	119
Caa	1146	42	781	323

and tends to decline as a company's credit rating declines.[5] The difference between the two hazard rates tends to increase as the credit rating declines.

Table 23.5 provides another way of looking at these results. It shows the excess return over the risk-free rate (still assumed to be the 7-year swap rate minus 10 basis points) earned by investors in bonds with different credit rating. Consider again an A-rated bond. The average spread over 7-year Treasuries is 111 basis points. Of this, 42 basis points are accounted for by the average spread between 7-year Treasuries and our proxy for the risk-free rate. A spread of 10 basis points is necessary to cover expected defaults. (This equals the historical hazard rate from Table 23.4 multiplied by 0.6 to allow for recoveries.) This leaves an excess return (after expected defaults have been taken into account) of 59 basis points.

Tables 23.4 and 23.5 show that a large percentage difference between default probability estimates translates into a small (but significant) excess return on the bond. For Aaa-rated bonds, the ratio of the two hazard rates is 17.0, but the expected excess return is only 34 basis points. The excess return tends to increase as credit quality declines.[6]

The excess return in Table 23.5 does not remain constant through time. Credit spreads, and therefore excess returns, were high in 2001, 2002, and the first half of 2003. After that they were fairly low until the credit crisis.

Real-World vs. Risk-Neutral Probabilities

The default probabilities implied from bond yields are risk-neutral probabilities of default. To explain why this is so, consider the calculations of default probabilities in Table 23.3. The calculations assume that expected default losses can be discounted at the risk-free rate. The risk-neutral valuation principle shows that this is a valid procedure providing the expected losses are calculated in a risk-neutral world. This means that the default probability Q in Table 23.3 must be a risk-neutral probability.

By contrast, the default probabilities implied from historical data are real-world default probabilities (sometimes also called *physical probabilities*). The expected excess return in Table 23.5 arises directly from the difference between real-world and risk-neutral default

[5] The results in Tables 23.4 and 23.5 are updates of the results in J. Hull, M. Predescu, and A. White, "Bond Prices, Default Probabilities, and Risk Premiums," *Journal of Credit Risk*, 1, 2 (Spring 2005): 53–60.

[6] The results for B-rated bonds in Tables 23.4 and 23.5 run counter to the overall pattern.

probabilities. If there were no expected excess return, then the real-world and risk-neutral default probabilities would be the same, and vice versa.

Why do we see such big differences between real-world and risk-neutral default probabilities? As we have just argued, this is the same as asking why corporate bond traders earn more than the risk-free rate on average.

One reason often advanced for the results is that corporate bonds are relatively illiquid and the returns on bonds are higher than they would otherwise be to compensate for this. This is true, but research shows that it does not fully explain the results in Table 23.5.[7] Another possible reason for the results is that the subjective default probabilities of bond traders may be much higher than the those given in Table 23.1. Bond traders may be allowing for depression scenarios much worse than anything seen during the period covered by historical data. However, it is difficult to see how this can explain a large part of the excess return that is observed.

By far the most important reason for the results in Tables 23.4 and 23.5 is that bonds do not default independently of each other. There are periods of time when default rates are very low and periods of time when they are very high. Evidence for this can be obtained by looking at the default rates in different years. Moody's statistics show that between 1970 and 2009 the default rate per year ranged from a low of 0.09% in 1979 to highs of 3.97% and 5.35% in 2001 and 2009, respectively. The year-to-year variation in default rates gives rise to systematic risk (i.e., risk that cannot be diversified away) and bond traders earn an excess expected return for bearing the risk. (This is similar to the excess expected return earned by equity holders that is calculated by the capital asset pricing model—see the appendix to Chapter 3.) The variation in default rates from year to year may be because of overall economic conditions and it may be because a default by one company has a ripple effect resulting in defaults by other companies. (The latter is referred to by researchers as *credit contagion.*)

In addition to the systematic risk we have just talked about there is nonsystematic (or idiosyncratic) risk associated with each bond. If we were talking about stocks, we would argue that investors can diversify the nonsystematic risk by choosing a portfolio of, say, 30 stocks. They should not therefore demand a risk premium for bearing nonsystematic risk. For bonds, the arguments are not so clear-cut. Bond returns are highly skewed with limited upside. (For example, on an individual bond, there might be a 99.75% chance of a 7% return in a year, and a 0.25% chance of a −60% return in the year, the first outcome corresponding to no default and the second to default.) This type of risk is difficult to "diversify away".[8] It would require tens of thousands of different bonds. In practice, many bond portfolios are far from fully diversified. As a result, bond traders may earn an extra return for bearing nonsystematic risk as well as for bearing the systematic risk mentioned in the previous paragraph.

Which Default Probability Estimate Should Be Used?

At this stage it is natural to ask whether we should use real-world or risk-neutral default probabilities in the analysis of credit risk. The answer depends on the purpose of the

[7] For example, J. Dick-Nielsen, P. Feldhütter, and D. Lando, "Corporate Bond Liquidity before and after the Onset of the Subprime Crisis," Working Paper, Copenhagen Business School, 2010, uses a number of different liquidity measures and a large database of bond trades. It shows that the liquidity component of credit spreads is relatively small.

[8] See J. D. Amato and E. M. Remolona, "The Credit Spread Puzzle," *BIS Quarterly Review*, 5 (Dec. 2003): 51–63.

analysis. When valuing credit derivatives or estimating the impact of default risk on the pricing of instruments, risk-neutral default probabilities should be used. This is because the analysis calculates the present value of expected future cash flows and almost invariably (implicitly or explicitly) involves using risk-neutral valuation. When carrying out scenario analyses to calculate potential future losses from defaults, real-world default probabilities should be used.

23.6 USING EQUITY PRICES TO ESTIMATE DEFAULT PROBABILITIES

When we use a table such as Table 23.1 to estimate a company's real-world probability of default, we are relying on the company's credit rating. Unfortunately, credit ratings are revised relatively infrequently. This has led some analysts to argue that equity prices can provide more up-to-date information for estimating default probabilities.

In 1974, Merton proposed a model where a company's equity is an option on the assets of the company.[9] Suppose, for simplicity, that a firm has one zero-coupon bond outstanding and that the bond matures at time T. Define:

V_0: Value of company's assets today
V_T: Value of company's assets at time T
E_0: Value of company's equity today
E_T: Value of company's equity at time T
D: Debt repayment due at time T
σ_V: Volatility of assets (assumed constant)
σ_E: Instantaneous volatility of equity.

If $V_T < D$, it is (at least in theory) rational for the company to default on the debt at time T. The value of the equity is then zero. If $V_T > D$, the company should make the debt repayment at time T and the value of the equity at this time is $V_T - D$. Merton's model, therefore, gives the value of the firm's equity at time T as

$$E_T = \max(V_T - D,\ 0)$$

This shows that the equity is a call option on the value of the assets with a strike price equal to the repayment required on the debt. The Black–Scholes–Merton formula gives the value of the equity today as

$$E_0 = V_0 N(d_1) - De^{-rT} N(d_2) \tag{23.3}$$

where

$$d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma_V\sqrt{T}$$

The value of the debt today is $V_0 - E_0$.

The risk-neutral probability that the company will default on the debt is $N(-d_2)$. To calculate this, we require V_0 and σ_V. Neither of these are directly observable. However, if the company is publicly traded, we can observe E_0. This means that equation (23.3) provides one condition that must be satisfied by V_0 and σ_V. We can also estimate σ_E

[9] See R. Merton "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates," *Journal of Finance*, 29 (1974): 449–70.

from historical data or options. From Itô's lemma,

$$\sigma_E E_0 = \frac{\partial E}{\partial V}\sigma_V V_0$$

or

$$\sigma_E E_0 = N(d_1)\sigma_V V_0 \tag{23.4}$$

This provides another equation that must be satisfied by V_0 and σ_V. Equations (23.3) and (23.4) provide a pair of simultaneous equations that can be solved for V_0 and σ_V.[10]

Example 23.1

The value of a company's equity is \$3 million and the volatility of the equity is 80%. The debt that will have to be paid in 1 year is \$10 million. The risk-free rate is 5% per annum. In this case $E_0 = 3$, $\sigma_E = 0.80$, $r = 0.05$, $T = 1$, and $D = 10$. Solving equations (23.3) and (23.4) yields $V_0 = 12.40$ and $\sigma_V = 0.2123$. The parameter d_2 is 1.1408, so that the probability of default is $N(-d_2) = 0.127$, or 12.7%. The market value of the debt is $V_0 - E_0$, or 9.40. The present value of the promised payment on the debt is $10e^{-0.05\times 1} = 9.51$. The expected loss on the debt is therefore $(9.51 - 9.40)/9.51$, or about 1.2% of its no-default value. The expected loss (EL) equals the probability of default (PD) times one minus the recovery rate. It follows that the recovery rate equals one minus EL/PD. In this case, the recovery rate is $1 - 1.2/12.7$, or about 91%, of the debt's no-default value.

The basic Merton model we have just presented has been extended in a number of ways. For example, one version of the model assumes that a default occurs whenever the value of the assets falls below a barrier level. Another allows payments on debt instruments to be required at more than one time.

How well do the default probabilities produced by Merton's model and its extensions correspond to actual default experience? The answer is that Merton's model and its extensions produce a good ranking of default probabilities (risk-neutral or real-world). This means that a monotonic transformation can be used to convert the probability of default output from Merton's model into a good estimate of either the real-world or risk-neutral default probability.[11] It may seem strange to take a default probability $N(-d_2)$ that is in theory a risk-neutral default probability (because it is calculated from an option-pricing model) and use it to estimate a real-world default probability. Given the nature of the calibration process we have just described, the underlying assumption is that the ranking of the risk-neutral default probabilities of different companies is the same as the ranking of their real-world default probabilities.

23.7 CREDIT RISK IN DERIVATIVES TRANSACTIONS

The credit exposure on a derivatives transaction is more complicated than that on a loan. This is because the claim that will be made in the event of a default is more uncertain. Consider a financial institution that has one derivatives contract outstanding

[10] To solve two nonlinear equations of the form $F(x, y) = 0$ and $G(x, y) = 0$, the Solver routine in Excel can be asked to find the values of x and y that minimize $[F(x, y)]^2 + [G(x, y)]^2$.

[11] Moody's KMV provides a service that transforms a default probability produced by Merton's model into a real-world default probability (which it refers to as an expected default frequency, or EDF). CreditGrades use Merton's model to estimate credit spreads, which are closely linked to risk-neutral default probabilities.

with a counterparty. Three possible situations can be distinguished:

1. Contract is always a liability to the financial institution
2. Contract is always an asset to the financial institution
3. Contract can become either an asset or a liability to the financial institution.

An example of a derivatives contract in the first category is a short option position; an example in the second category is a long option position; an example in the third category is a forward contract.

Derivatives in the first category have no credit risk to the financial institution. If the counterparty goes bankrupt, there will be no loss. The derivative is one of the counterparty's assets. It is likely to be retained, closed out, or sold to a third party. The result is no loss (or gain) to the financial institution.

Derivatives in the second category always have credit risk to the financial institution. If the counterparty goes bankrupt, a loss is likely to be experienced. The derivative is one of the counterparty's liabilities. The financial institution has to make a claim against the assets of the counterparty and may receive some percentage of the value of the derivative. (Typically, a claim arising from a derivatives transaction is unsecured and junior.)

Derivatives in the third category may or may not have credit risk. If the counterparty defaults when the value of the derivative is positive to the financial institution, a claim will be made against the assets of the counterparty and a loss is likely to be experienced. If the counterparty defaults when the value is negative to the financial institution, no loss is made because the derivative is retained, closed out, or sold to a third party.[12]

Adjusting Derivatives' Valuations for Counterparty Default Risk

How should a financial institution (or end-user of derivatives) adjust the value of a derivative to allow for counterparty credit risk? Consider a derivative that lasts until time T and has a value of f_0 today assuming no defaults. Let us suppose that defaults can take place at times $t_1, t_2, \ldots, t_n$, where $t_n = T$, and that the value of the derivative to the financial institution (assuming no defaults) at time t_i is f_i. Define the risk-neutral probability of default at time t_i as q_i and the expected recovery rate as R.[13]

The exposure at time t_i is the financial institution's potential loss. This is $\max(f_i, 0)$. Assume that the expected recovery in the event of a default is R times the exposure. Assume also that the recovery rate and the probability of default are independent of the value of the derivative. The risk-neutral expected loss from default at time t_i is

$$q_i(1 - R)\hat{E}[\max(f_i, 0)]$$

where $\hat{E}$ denotes expected value in a risk-neutral world. Taking present values leads to what is termed the *credit value adjustment* (CVA):

$$\sum_{i=1}^{n} u_i\, v_i \tag{23.5}$$

where u_i equals $q_i(1 - R)$ and v_i is the value today of an instrument that pays off the exposure on the derivative under consideration at time t_i.

[12] Note that a company usually defaults because of a deterioration in its overall financial health, not because of the value of any one transaction.

[13] The probability of default could be calculated from bond prices in the way described in Section 23.4.

Consider again the three categories of derivatives mentioned earlier. The first category (where the derivative is always a liability to the financial institution) is easy to deal with. The value of f_i is always negative and so the total expected loss from defaults given by equation (23.5) is always zero. The financial institution needs to make no adjustments for the cost of defaults. (Of course, the counterparty may want to take account of the possibility of the financial institution defaulting in its own pricing.)

For the second category (where the derivative is always an asset to the financial institution), f_i is always positive. This means than the expression $\max(f_i, 0)$ always equals f_i. Suppose that the only payoff from the derivative is at time T, the end of its life. In this case, f_0 must be the present value of f_i, so that $v_i = f_0$ for all i. The expression in equation (23.5) for the present value of the cost of defaults becomes

$$f_0 \sum_{i=1}^{n} q_i(1 - R)$$

If f_0^* is the actual value of the derivative (after allowing for possible defaults), then

$$f_0^* = f_0 - f_0 \sum_{i=1}^{n} q_i(1 - R) = f_0\left[1 - \sum_{i=1}^{n} q_i(1 - R)\right] \tag{23.6}$$

One particular instrument that falls into the second category we are considering is an unsecured zero-coupon bond that promises \$1 at time T and is issued by the counterparty in the derivatives transaction. Define B_0 as the value of the bond assuming no possibility of default and B_0^* as the actual value of the bond. If we make the simplifying assumption that the recovery on the bond as a percent of its no-default value is the same as that on the derivative, then

$$B_0^* = B_0\left[1 - \sum_{i=1}^{n} q_i(1 - R)\right] \tag{23.7}$$

From equations (23.6) and (23.7),

$$\frac{f_0^*}{f_0} = \frac{B_0^*}{B_0} \tag{23.8}$$

If y is the yield on a risk-free zero-coupon bond maturing at time T and y^* is the yield on a zero-coupon bond issued by the counterparty that matures at time T, then $B_0 = e^{-yT}$ and $B_0^* = e^{-y^*T}$, so that equation (23.8) gives

$$f_0^* = f_0 e^{-(y^* - y)T} \tag{23.9}$$

This shows that any derivative promising a payoff at time T can be valued by increasing the discount rate that is applied to the expected payoff in a risk-neutral world from the risk-free rate y to the risky rate y^*.

Example 23.2

Consider a 2-year over-the-counter option sold by company X with a value, assuming no possibility of default, of \$3. Suppose that 2-year zero-coupon bonds issued by the company X have a yield that is 1.5% greater than a similar risk-free zero-coupon bond. The value of the option is $3e^{-0.015\times 2} = 2.91$, or \$2.91.

For the third category of derivatives, the sign of f_i is uncertain. The variable v_i is a call option on f_i with a strike price of zero. One way of calculating v_i is to simulate the underlying market variables over the life of the derivative. Sometimes approximate analytic calculations are possible (see, e.g., Problems 23.15 and 23.16).

The analyses we have presented assume that the probability of default is independent of the value of the derivative. This is likely to be a reasonable approximation in circumstances when the derivative is a small part of the portfolio of the counterparty or when the counterparty is using the derivative for hedging purposes. When a counterparty wants to enter into a large derivatives transaction for speculative purposes a financial institution should be wary. When the transaction has a large negative value for the counterparty (and a large positive value for the financial institution), the chance of counterparty declaring bankruptcy may be much higher than when the situation is the other way round.

Traders working for a financial institution use the term *right-way risk* to describe the situation where a counterparty is most likely to default when the financial institution has zero, or very little, exposure. They use the term *wrong-way risk* to describe the situation where the counterparty is most likely to default when the financial institution has a big exposure.

23.8 CREDIT RISK MITIGATION

In many instances the analysis we just have presented overstates the credit risk in a derivatives transaction. This is because there are a number of clauses that derivatives dealers include in their contracts to mitigate credit risk.

Netting

A clause that has become standard in the Master Agreements that govern transactions in the over-the-counter market is known as *netting*. This states that, if a company defaults on one transaction it has with a counterparty, it must default on all outstanding transactions with the counterparty.

Netting has been successfully tested in the courts in most jurisdictions. It can substantially reduce credit risk for a financial institution. Consider, for example, a financial institution that has three transactions outstanding with a particular counterparty. The transactions are worth +\$10 million, +\$30 million, and −\$25 million to the financial institution. Suppose the counterparty runs into financial difficulties and defaults on its outstanding obligations. To the counterparty, the three transactions have values of −\$10 million, −\$30 million, and +\$25 million, respectively. Without netting, the counterparty would default on the first two transactions and retain the third for a loss to the financial institution of \$40 million. With netting, it is compelled to default on all three transactions for a loss to the financial institution of \$15 million.[14]

Suppose a financial institution has a portfolio of N derivatives transactions with a particular counterparty. Suppose that the no-default value of the ith transaction is V_i

[14] Note that, if the third transaction were worth −\$45 million to the financial institution instead of −\$25 million, the counterparty would choose not to default and there would be no loss to the financial institution.

and the amount recovered in the event of default is the recovery rate times this no default value. Without netting, the financial institution loses

$$(1 - R)\sum_{i=1}^{N} \max(V_i, 0)$$

where R is the recovery rate. With netting, it loses

$$(1 - R)\max\left(\sum_{i=1}^{N} V_i, 0\right)$$

Without netting, its loss is the payoff from a portfolio of call options on the transactions where each option has a strike price of zero. With netting, it is the payoff from a single option on the portfolio of transactions with a strike price of zero. The value of an option on a portfolio is never greater than, and is often considerably less than, the value of the corresponding portfolio of options.

The CVA analysis presented in the previous section can be extended so that equation (23.5) gives the present value of the expected loss from all transactions with a counterparty when netting agreements are in place. This is achieved by redefining v_i in the equation as the present value of a derivative that pays off the exposure at time t_i on the portfolio of all transactions with a counterparty.

A challenging task for a financial institution when considering whether it should enter into a new derivatives transaction with a counterparty is to calculate the incremental effect on expected credit losses. This can be done by using equation (23.5) in the way just described to calculate expected default costs with and without the transaction. It is interesting to note that, because of netting, the incremental effect of a new transaction on expected default losses can be negative. This happens when the value of the new transaction is negatively correlated with the value of existing transactions.

Collateralization

Another clause frequently used to mitigate credit risks is known as *collateralization*. Suppose that a company and a financial institution have entered into a number of derivatives transactions. A typical collateralization agreement specifies that the transactions be valued periodically. If the total value of the transactions to the financial institution is above a specified threshold level, the agreement requires the cumulative collateral posted by the company to equal the difference between the value of the transactions to the financial institution and the threshold level. If, after the collateral has been posted, the value of the transactions moves in favor of the company so that the difference between value of the transactions to the financial institution and the threshold level is less than the total margin already posted, the company can reclaim margin. In the event of a default by the company, the financial institution can seize the collateral. If the company does not post collateral as required, the financial institution can close out the transactions.

Suppose, for example, that the threshold level for the company is \$10 million and the transactions are marked to market daily for the purposes of collateralization. If on a particular day the value of the transactions to the financial institution rises from \$9 million to \$10.5 million, it can ask for \$0.5 million of collateral. If the next day the value of the transactions rises further to \$11.4 million, it can ask for a further \$0.9 million of collateral. If the value of the transactions falls to \$10.9 million on the following day, the company can ask for \$0.5 million of the collateral to be returned. Note that the threshold (\$10 million in this case) can be regarded as a line of credit that the financial institution is prepared to grant to the company.

The margin must be deposited by the company with the financial institution in cash or in the form of acceptable securities such as bonds. The securities are subject to a discount known as a *haircut* applied to their market value for the purposes of margin calculations. Interest is normally paid on cash.

If the collateralization agreement is a two-way agreement a threshold will also be specified for the financial institution. The company can then ask the financial institution to post collateral when the value of the outstanding contracts to the company exceeds the threshold.

Collateralization agreements provide a great deal of protection against the possibility of default (just as the margin accounts discussed in Chapter 2 provide protection for people who trade futures on an exchange). However, the threshold amount is not subject to protection. Furthermore, even when the threshold is zero, the protection is not total. This is because, when a company gets into financial difficulties, it is likely to stop responding to requests to post collateral. By the time the counterparty exercises its right to close out contracts, their value may have moved further in its favor.

As explained in Chapter 2, over-the-counter derivatives are increasingly moving to clearing houses where market participants post both an initial margin and maintenance margins.

Downgrade Triggers

Another credit risk mitigation technique sometimes used by a financial institution is known as a *downgrade trigger*. This is a clause stating that if the credit rating of the counterparty falls below a certain level, say Baa, the financial institution has the option to close out a derivatives transaction at its market value.

Downgrade triggers do not provide protection from a big jump in a company's credit rating (for example, from A to default). Also, downgrade triggers work well only if relatively little use is made of them. If a company has many downgrade triggers outstanding with its counterparties, they are liable to provide little protection to any of the counterparties (see Business Snapshot 23.1).

23.9 DEFAULT CORRELATION

The term *default correlation* is used to describe the tendency for two companies to default at about the same time. There are a number of reasons why default correlation exists. Companies in the same industry or the same geographic region tend to be affected similarly by external events and as a result may experience financial difficulties at the same time. Economic conditions generally cause average default rates to be higher in some years than in other years. A default by one company may cause a default by another—the credit contagion effect. Default correlation means that credit risk cannot be completely diversified away and is the major reason why risk-neutral default probabilities are greater than real-world default probabilities (see Section 23.5).

Business Snapshot 23.1 Downgrade Triggers and Enron's Bankruptcy

In December 2001, Enron, one of the largest companies in the United States, went bankrupt. Right up to the last few days, it had an investment-grade credit rating. The Moody's rating immediately prior to default was Baa3 and the S&P rating was BBB–. The default was, however, anticipated to some extent by the stock market because Enron's stock price fell sharply in the period leading up to the bankruptcy. The probability of default estimated by models such as the one described in Section 23.6 increased sharply during this period.

Enron had entered into a huge number of derivatives transactions with downgrade triggers. The downgrade triggers stated that, if its credit rating fell below investment grade (i.e., below Baa3/BBB–), its counterparties would have the option of closing out the transactions. Suppose that Enron had been downgraded to below investment grade in, say, October 2001. The transactions that counterparties would choose to close out would be those with negative values to Enron (and positive values to the counterparties). So, Enron would have been required to make huge cash payments to its counterparties. It would not have been able to do this and immediate bankruptcy would have resulted.

This example illustrates that downgrade triggers provide protection only when relatively little use is made of them. When a company enters into a huge number of contracts with downgrade triggers, they may actually cause a company to go bankrupt prematurely. In Enron's case, we could argue that it was going to go bankrupt anyway and accelerating the event by two months would not have done any harm. In fact, Enron did have a chance of survival in October 2001. Attempts were being made to work out a deal with another energy company, Dynergy, and so forcing bankruptcy in October 2001 was not in the interests of either creditors or shareholders.

The credit rating companies found themselves in a difficult position. If they downgraded Enron to recognize its deteriorating financial position, they were signing its death warrant. If they did not do so, there was a chance of Enron surviving.

Default correlation is important in the determination of probability distributions for default losses from a portfolio of exposures to different counterparties.[15] Two types of default correlation models that have been suggested by researchers are referred to as *reduced form models* and *structural models*.

Reduced form models assume that the hazard rates for different companies follow stochastic processes and are correlated with macroeconomic variables. When the hazard rate for company A is high there is a tendency for the hazard rate for company B to be high. This induces a default correlation between the two companies.

Reduced form models are mathematically attractive and reflect the tendency for economic cycles to generate default correlations. Their main disadvantage is that the range of default correlations that can be achieved is limited. Even when there is a perfect correlation between the hazard rates of the two companies, the probability that they will both default during the same short period of time is usually very low. This is liable to be a problem in some circumstances. For example, when two companies operate in the same

[15] A binomial correlation measure that has been used by rating agencies is described in Technical Note 26 at www.rotman.utoronto.ca/~hull/TechnicalNotes.

industry and the same country or when the financial health of one company is for some reason heavily dependent on the financial health of another company, a relatively high default correlation may be warranted. One approach to solving this problem is by extending the model so that the hazard rate exhibits large jumps.

Structural models are based on a model similar to Merton's model (see Section 23.6). A company defaults if the value of its assets is below a certain level. Default correlation between companies A and B is introduced into the model by assuming that the stochastic process followed by the assets of company A is correlated with the stochastic process followed by the assets of company B. Structural models have the advantage over reduced form models that the correlation can be made as high as desired. Their main disadvantage is that they are liable to be computationally quite slow.

The Gaussian Copula Model for Time to Default

A model that has become a popular practical tool is the Gaussian copula model for the time to default. It can be characterized as a simplified structural model. It assumes that all companies will default eventually and attempts to quantify the correlation between the probability distributions of the times to default for two or more different companies.

The model can be used in conjunction with either real-world or risk-neutral default probabilities. The left tail of the real-world probability distribution for the time to default of a company can be estimated from data produced by rating agencies such as that in Table 23.1. The left tail of the risk-neutral probability distribution of the time to default can be estimated from bond prices using the approach in Section 23.4.

Define t_1 as the time to default of company 1 and t_2 as the time to default of company 2. If the probability distributions of t_1 and t_2 were normal, we could assume that the joint probability distribution of t_1 and t_2 is bivariate normal. As it happens, the probability distribution of a company's time to default is not even approximately normal. This is where a Gaussian copula model comes in. We transform t_1 and t_2 into new variables x_1 and x_2 using

$$x_1 = N^{-1}[Q_1(t_1)], \qquad x_2 = N^{-1}[Q_2(t_2)]$$

where Q_1 and Q_2 are the cumulative probability distributions for t_1 and t_2, respectively, and N^{-1} is the inverse of the cumulative normal distribution ($u = N^{-1}(v)$ when $v = N(u)$). These are "percentile-to-percentile" transformations. The 5-percentile point in the probability distribution for t_1 is transformed to $x_1 = -1.645$, which is the 5-percentile point in the standard normal distribution; the 10-percentile point in the probability distribution for t_1 is transformed to $x_1 = -1.282$, which is the 10-percentile point in the standard normal distribution, and so on. The t_2-to-x_2 transformation is similar.

By construction, x_1 and x_2 have normal distributions with mean zero and unit standard deviation. The model assumes that the joint distribution of x_1 and x_2 is bivariate normal. This assumption is referred to as using a *Gaussian copula*. The assumption is convenient because it means that the joint probability distribution of t_1 and t_2 is fully defined by the cumulative default probability distributions Q_1 and Q_2 for t_1 and t_2, together with a single correlation parameter.

The attraction of the Gaussian copula model is that it can be extended to many companies. Suppose that we are considering n companies and that t_i is the time to default of the ith company. We transform each t_i into a new variable, x_i, that has a standard normal distribution. The transformation is the percentile-to-percentile transformation

$$x_i = N^{-1}[Q_i(t_i)]$$

where Q_i is the cumulative probability distribution for t_i. It is then assumed that the x_i are multivariate normal. The default correlation between t_i and t_j is measured as the correlation between x_i and x_j. This is referred to as the *copula correlation*.[16]

The Gaussian copula is a useful way of representing the correlation structure between variables that are not normally distributed. It allows the correlation structure of the variables to be estimated separately from their marginal (unconditional) distributions. Although the variables themselves are not multivariate normal, the approach assumes that after a transformation is applied to each variable they are multivariate normal.

Example 23.3

Suppose that we wish to simulate defaults during the next 5 years in 10 companies. The copula default correlations between each pair of companies is 0.2. For each company the cumulative probability of a default during the next 1, 2, 3, 4, 5 years is 1%, 3%, 6%, 10%, 15%, respectively. When a Gaussian copula is used we sample from a multivariate normal distribution to obtain the x_i $(1 \leqslant i \leqslant 10)$ with the pairwise correlation between the x_i being 0.2. We then convert the x_i to t_i, a time to default. When the sample from the normal distribution is less than $N^{-1}(0.01) = -2.33$, a default takes place within the first year; when the sample is between -2.33 and $N^{-1}(0.03) = -1.88$, a default takes place during the second year; when the sample is between -1.88 and $N^{-1}(0.06) = -1.55$, a default takes place during the third year; when the sample is between -1.55 and $N^{-1}(0.10) = -1.28$, a default takes place during the fourth year; when the sample is between -1.28 and $N^{-1}(0.15) = -1.04$, a default takes place during the fifth year. When the sample is greater than -1.04, there is no default during the 5 years.

A Factor-Based Correlation Structure

To avoid defining a different correlation between x_i and x_j for each pair of companies i and j in the Gaussian copula model, a one-factor model is often used. The assumption is that

$$x_i = a_i F + \sqrt{1 - a_i^2}\, Z_i \tag{23.10}$$

In this equation, F is a common factor affecting defaults for all companies and Z_i is a factor affecting only company i. The variable F and the variables Z_i have independent standard normal distributions. The a_i are constant parameters between -1 and $+1$. The correlation between x_i and x_j is $a_i a_j$.[17]

Suppose that the probability that company i will default by a particular time T is $Q_i(T)$. Under the Gaussian copula model, a default happens by time T when

[16] As an approximation, the copula correlation between t_i and t_j is often assumed to be the correlation between the equity returns for companies i and j.

[17] The parameter a_i is sometimes approximated as the correlation of company i's equity returns with a well-diversified market index.

$N(x_i) < Q_i(T)$ or $x_i < N^{-1}[Q_i(T)]$. From equation (23.10), this condition is

$$a_i F + \sqrt{1 - a_i^2}\, Z_i < N^{-1}[Q_i(T)]$$

or

$$Z_i < \frac{N^{-1}[Q_i(T)] - a_i F}{\sqrt{1 - a_i^2}}$$

Conditional on the value of the factor F, the probability of default is therefore

$$Q_i(T \mid F) = N\left(\frac{N^{-1}[Q_i(T)] - a_i F}{\sqrt{1 - a_i^2}}\right) \tag{23.11}$$

A particular case of the one-factor Gaussian copula model is where the probability distributions of default are the same for all i and the correlation between x_i and x_j is the same for all i and j. Suppose that $Q_i(T) = Q(T)$ for all i and that the common correlation is ρ, so that $a_i = \sqrt{\rho}$ for all i. Equation (23.11) becomes

$$Q(T \mid F) = N\left(\frac{N^{-1}[Q(T)] - \sqrt{\rho}\, F}{\sqrt{1 - \rho}}\right) \tag{23.12}$$

23.10 CREDIT VaR

Credit value at risk can be defined analogously to the way value at risk is defined for market risks (see Chapter 21). For example, a credit VaR with a confidence level of 99.9% and a 1-year time horizon is the credit loss that we are 99.9% confident will not be exceeded over 1 year.

Consider a bank with a very large portfolio of similar loans. As an approximation, assume that the probability of default is the same for each loan and the correlation between each pair of loans is the same. When the Gaussian copula model for time to default is used, the right-hand side of equation (23.12) is to a good approximation equal to the percentage of defaults by time T as a function of F. The factor F has a standard normal distribution. We are X% certain that its value will be greater than $N^{-1}(1 - X) = -N^{-1}(X)$. We are therefore X% certain that the percentage of losses over T years on a large portfolio will be less than $V(X, T)$, where

$$V(X, T) = N\left(\frac{N^{-1}[Q(T)] + \sqrt{\rho}\, N^{-1}(X)}{\sqrt{1 - \rho}}\right) \tag{23.13}$$

This result was first produced by Vasicek.[18] As in equation (23.12), $Q(T)$ is the probability of default by time T and ρ is the copula correlation between any pair of loans.

A rough estimate of the credit VaR when an X% confidence level is used and the time horizon is T is therefore $L(1 - R)V(X, T)$, where L is the size of the loan portfolio and R is the recovery rate. The contribution of a particular loan of size L_i to the credit VaR

[18] See O. Vasicek, "Probability of Loss on a Loan Portfolio," Working Paper, KMV, 1987. Vasicek's results were published in *Risk* magazine in December 2002 under the title "Loan Portfolio Value".

is $L_i(1 - R)V(X, T)$. This model underlies some of the formulas that regulators use for credit risk capital.[19]

Example 23.4

Suppose that a bank has a total of \$100 million of retail exposures. The 1-year probability of default averages 2% and the recovery rate averages 60%. The copula correlation parameter is estimated as 0.1. In this case,

$$V(0.999, 1) = N\left(\frac{N^{-1}(0.02) + \sqrt{0.1}\,N^{-1}(0.999)}{\sqrt{1-0.1}}\right) = 0.128$$

showing that the 99.9% worst case default rate is 12.8%. The 1-year 99.9% credit VaR is therefore $100 \times 0.128 \times (1 - 0.6)$ or \$5.13 million.

CreditMetrics

Many banks have developed other procedures for calculating credit VaR for internal use. One popular approach is known as CreditMetrics. This involves estimating a probability distribution of credit losses by carrying out a Monte Carlo simulation of the credit rating changes of all counterparties. Suppose we are interested in determining the probability distribution of losses over a 1-year period. On each simulation trial, we sample to determine the credit rating changes and defaults of all counterparties during the year. We then revalue our outstanding contracts to determine the total of credit losses for the year. After a large number of simulation trials, a probability distribution for credit losses is obtained. This can be used to calculate credit VaR.

This approach is liable to be computationally quite time intensive. However, it has the advantage that credit losses are defined as those arising from credit downgrades as well as defaults. Also the impact of credit mitigation clauses such as those described in Section 23.8 can be approximately incorporated into the analysis.

Table 23.6 is typical of the historical data provided by rating agencies on credit rating changes and could be used as a basis for a CreditMetrics Monte Carlo simulation. It shows the percentage probability of a bond moving from one rating category to another during a 1-year period. For example, a bond that starts with an A credit rating has a 90.91% chance of still having an A rating at the end of 1 year. It has a 0.05% chance of defaulting during the year, a 0.09% chance of dropping to B, and so on.[20]

In sampling to determine credit losses, the credit rating changes for different counterparties should not be assumed to be independent. A Gaussian copula model is typically used to construct a joint probability distribution of rating changes similarly to the way it is used in the model in the previous section to describe the joint probability distribution of times to default. The copula correlation between the rating transitions for two companies is usually set equal to the correlation between their equity returns using a factor model similar to that in Section 23.9.

As an illustration of the CreditMetrics approach suppose that we are simulating the rating change of a Aaa and a Baa company over a 1-year period using the transition

[19] For more details, see J. Hull, *Risk Management and Financial Institutions*, 2nd edn. Upper Saddle River, Pearson, 2010.

[20] Technical Note 11 at www.rotman.utoronto.ca/~hull/TechnicalNotes explains how a table such as Table 23.6 can be used to calculate transition matrices for periods other than 1 year.

Table 23.6 One-year ratings transition matrix, 1970–2009, with probabilities expressed as percentages and adjustments for transitions to the WR (without rating) category. *Source*: Moody's.

Initial rating	*Rating at year-end*								
	Aaa	*Aa*	*A*	*Baa*	*Ba*	*B*	*Caa*	*Ca–C*	*Default*
Aaa	90.57	8.76	0.63	0.01	0.03	0.00	0.00	0.00	0.00
Aa	1.06	90.30	8.19	0.36	0.05	0.02	0.01	0.00	0.02
A	0.06	2.90	90.91	5.44	0.50	0.09	0.03	0.00	0.05
Baa	0.04	0.20	4.91	89.18	4.44	0.83	0.19	0.02	0.17
Ba	0.01	0.07	0.42	6.24	83.47	7.99	0.58	0.09	1.13
B	0.01	0.04	0.15	0.39	5.40	82.50	6.35	0.79	4.37
Caa	0.00	0.02	0.02	0.19	0.51	9.55	70.01	4.97	14.72
Ca-C	0.00	0.00	0.00	0.00	0.43	3.01	11.61	51.67	33.28
Default	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

matrix in Table 23.6. Suppose that the correlation between the equities of the two companies is 0.2. On each simulation trial, we would sample two variables x_A and x_B from normal distributions so that their correlation is 0.2. The variable x_A determines the new rating of the Aaa company and variable x_B determines the new rating of the Baa company. Since $N^{-1}(0.9057) = 1.3147$, the Aaa company stays Aaa if $x_A < 1.3147$; since $N^{-1}(0.9057 + 0.0876) = 2.4730$, it becomes Aa if $1.3147 \leqslant x_A < 2.4730$; since $N^{-1}(0.9057 + 0.0876 + 0.0063) = 3.3528$, it becomes A if $2.4730 \leqslant x_A < 3.3528$; and so on. Consider next the Baa company. Since $N^{-1}(0.0004) = -3.3528$, the Baa company becomes Aaa if $x_B < -3.3528$; since $N^{-1}(0.0004 + 0.0020) = -2.8202$, it becomes Aa if $-3.3528 \leqslant x_B < -2.8202$; since

$$N^{-1}(0.0004 + 0.0020 + 0.0491) = -1.6305$$

it becomes A if $-2.8202 \leqslant x_B < -1.6305$; and so on. The Aaa never defaults during the year. The Baa defaults when $x_B > N^{-1}(0.9983)$, that is when $x_B > 2.9290$.

SUMMARY

The probability that a company will default during a particular period of time in the future can be estimated from historical data, bond prices, or equity prices. The default probabilities calculated from bond prices are risk-neutral probabilities, whereas those calculated from historical data are real-world probabilities. Real-world probabilities should be used for scenario analysis and the calculation of credit VaR. Risk-neutral probabilities should be used for valuing credit-sensitive instruments. Risk-neutral default probabilities are often significantly higher than real-world default probabilities.

The expected loss experienced from a counterparty default is reduced by what is known as netting. This is a clause in most contracts written by a financial institution stating that, if a counterparty defaults on one contract it has with the financial institution, it must default on all contracts it has with the financial institution. Losses are also reduced by collateralization and downgrade triggers. Collateralization requires the counterparty to post collateral and a downgrade trigger gives a financial institution the option to close out a contract if the credit rating of a counterparty falls below a specified level.

Credit VaR can be defined similarly to the way VaR is defined for market risk. One approach to calculating it is the Gaussian copula model of time to default. This is used by regulators in the calculation of capital for credit risk. Another popular approach for calculating credit VaR is CreditMetrics. This uses a Gaussian copula model for credit rating changes.

FURTHER READING

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Practice Questions (Answers in the Solutions Manual)

23.1. The spread between the yield on a 3-year corporate bond and the yield on a similar risk-free bond is 50 basis points. The recovery rate is 30%. Estimate the average hazard rate per year over the 3-year period.

23.2. Suppose that in Problem 23.1 the spread between the yield on a 5-year bond issued by the same company and the yield on a similar risk-free bond is 60 basis points. Assume the same recovery rate of 30%. Estimate the average hazard rate per year over the 5-year period. What do your results indicate about the average hazard rate in years 4 and 5?

23.3. Should researchers use real-world or risk-neutral default probabilities for (a) calculating credit value at risk and (b) adjusting the price of a derivative for defaults?

23.4. How are recovery rates usually defined?

23.5. Explain the difference between an unconditional default probability density and a hazard rate.

23.6. Verify (a) that the numbers in the second column of Table 23.4 are consistent with the numbers in Table 23.1 and (b) that the numbers in the fourth column of Table 23.5 are consistent with the numbers in Table 23.4 and a recovery rate of 40%.

23.7. Describe how netting works. A bank already has one transaction with a counterparty on its books. Explain why a new transaction by a bank with a counterparty can have the effect of increasing or reducing the bank's credit exposure to the counterparty.

23.8. What is meant by a "haircut" in a collateralization agreement. A company offers to post its own equity as collateral. How would you respond?

23.9. Explain the difference between the Gaussian copula model for the time to default and CreditMetrics as far as the following are concerned: (a) the definition of a credit loss and (b) the way in which default correlation is modeled.

23.10. Suppose that the LIBOR/swap curve is flat at 6% with continuous compounding and a 5-year bond with a coupon of 5% (paid semiannually) sells for 90.00. How would an asset swap on the bond be structured? What is the asset swap spread that would be calculated in this situation?

23.11. Show that the value of a coupon-bearing corporate bond is the sum of the values of its constituent zero-coupon bonds when the amount claimed in the event of default is the no-default value of the bond, but that this is not so when the claim amount is the face value of the bond plus accrued interest.

23.12. A 4-year corporate bond provides a coupon of 4% per year payable semiannually and has a yield of 5% expressed with continuous compounding. The risk-free yield curve is flat at 3% with continuous compounding. Assume that defaults can take place at the end of each year (immediately before a coupon or principal payment) and that the recovery rate is 30%. Estimate the risk-neutral default probability on the assumption that it is the same each year.

23.13. A company has issued 3- and 5-year bonds with a coupon of 4% per annum payable annually. The yields on the bonds (expressed with continuous compounding) are 4.5% and 4.75%, respectively. Risk-free rates are 3.5% with continuous compounding for all maturities. The recovery rate is 40%. Defaults can take place halfway through each year. The risk-neutral default rates per year are Q_1 for years 1 to 3 and Q_2 for years 4 and 5. Estimate Q_1 and Q_2.

23.14. Suppose that a financial institution has entered into a swap dependent on the sterling interest rate with counterparty X and an exactly offsetting swap with counterparty Y. Which of the following statements are true and which are false?

(a) The total present value of the cost of defaults is the sum of the present value of the cost of defaults on the contract with X plus the present value of the cost of defaults on the contract with Y.
(b) The expected exposure in 1 year on both contracts is the sum of the expected exposure on the contract with X and the expected exposure on the contract with Y.
(c) The 95% upper confidence limit for the exposure in 1 year on both contracts is the sum of the 95% upper confidence limit for the exposure in 1 year on the contract with X and the 95% upper confidence limit for the exposure in 1 year on the contract with Y.

Explain your answers.

23.15. A company enters into a 1-year forward contract to sell \$100 for AUD150. The contract is initially at the money. In other words, the forward exchange rate is 1.50. The 1-year dollar risk-free rate of interest is 5% per annum. The 1-year dollar rate of interest at which the counterparty can borrow is 6% per annum. The exchange rate volatility is 12% per annum. Estimate the present value of the cost of defaults on the contract. Assume that defaults are recognized only at the end of the life of the contract.

23.16. Suppose that in Problem 23.15, the 6-month forward rate is also 1.50 and the 6-month dollar risk-free interest rate is 5% per annum. Suppose further that the 6-month dollar rate of interest at which the counterparty can borrow is 5.5% per annum. Estimate the present value of the cost of defaults assuming that defaults can occur either at the 6-month point or at the 1-year point? (If a default occurs at the 6-month point, the company's potential loss is the market value of the contract.)

23.17. "A long forward contract subject to credit risk is a combination of a short position in a no-default put and a long position in a call subject to credit risk." Explain this statement.

23.18. Explain why the credit exposure on a matched pair of forward contracts resembles a straddle.

23.19. Explain why the impact of credit risk on a matched pair of interest rate swaps tends to be less than that on a matched pair of currency swaps.

23.20. "When a bank is negotiating currency swaps, it should try to ensure that it is receiving the lower interest rate currency from a company with a low credit risk." Explain why.

23.21. Does put–call parity hold when there is default risk? Explain your answer.

23.22. Suppose that in an asset swap B is the market price of the bond per dollar of principal, B^* is the default-free value of the bond per dollar of principal, and V is the present value of the asset swap spread per dollar of principal. Show that $V = B^* - B$.

23.23. Show that under Merton's model in Section 23.6 the credit spread on a T-year zero-coupon bond is $-\ln[N(d_2) + N(-d_1)/L]/T$, where $L = De^{-rT}/V_0$.

23.24. Suppose that the spread between the yield on a 3-year zero-coupon riskless bond and a 3-year zero-coupon bond issued by a corporation is 1%. By how much does Black–Scholes–Merton overstate the value of a 3-year European option sold by the corporation.

23.25. Give an example of (a) right-way risk and (b) wrong-way risk.

Further Questions

23.26. Suppose a 3-year corporate bond provides a coupon of 7% per year payable semiannually and has a yield of 5% (expressed with semiannual compounding). The yields for all maturities on risk-free bonds is 4% per annum (expressed with semiannual compounding). Assume that defaults can take place every 6 months (immediately before a coupon payment) and the recovery rate is 45%. Estimate the default probabilities assuming (a) that the unconditional default probabilities are the same on each possible default date and (b) that the default probabilities conditional on no earlier default are the same on each possible default date.

23.27. A company has 1- and 2-year bonds outstanding, each providing a coupon of 8% per year payable annually. The yields on the bonds (expressed with continuous compounding) are

6.0% and 6.6%, respectively. Risk-free rates are 4.5% for all maturities. The recovery rate is 35%. Defaults can take place halfway through each year. Estimate the risk-neutral default rate each year.

23.28. Explain carefully the distinction between real-world and risk-neutral default probabilities. Which is higher? A bank enters into a credit derivative where it agrees to pay $100 at the end of 1 year if a certain company's credit rating falls from A to Baa or lower during the year. The 1-year risk-free rate is 5%. Using Table 23.6, estimate a value for the derivative. What assumptions are you making? Do they tend to overstate or understate the value of the derivative.

23.29. The value of a company's equity is $4 million and the volatility of its equity is 60%. The debt that will have to be repaid in 2 years is $15 million. The risk-free interest rate is 6% per annum. Use Merton's model to estimate the expected loss from default, the probability of default, and the recovery rate in the event of default. (*Hint*: The Solver function in Excel can be used for this question, as indicated in footnote 10.)

23.30. Suppose that a bank has a total of $10 million of exposures of a certain type. The 1-year probability of default averages 1% and the recovery rate averages 40%. The copula correlation parameter is 0.2. Estimate the 99.5% 1-year credit VaR.